

This point of view explores that shift and what it means for original equipment manufacturers (OEMs), for embedded systems design, and for the rise of a new Tier-0.5 model, where suppliers evolve from component providers into strategic partners shaping intelligence, interaction, and experience at the edge.

Executive summary

Electronics everywhere. Electronic components today may account for more than ~35% of the price of an automobile. By 2030, this could approach 50%, turning even simple products into smart devices. This surge in electronic content means even traditionally simple products are becoming ‘smart devices,’ blurring the line between mechanical parts and digital technology.

Edge computing and distributed intelligence. Instead of routing all data to a central brain (ECU) or the cloud, there is a shift towards edge computing at a local or ‘zonal’ level. In vehicles, this is seen in zonal architectures where each zone of the car has its own controller for local sensors/actuators. This distributed intelligence is powered by new microcontrollers and systems-on-chips (SoCs) that can run tiny machine learning (TinyML) locally, enabling AI within the small, low-powered part itself.

AI-integrated microcontrollers (MCUs). Advances in silicon are allowing tiny ML models on MCUs to let devices interpret sensor data and adapt without needing cloud connectivity.

Embedded HMI and micro interactions. As intelligence decentralises, HMI is also moving to the edge. An intelligent subsystem or part now often needs its own means of communicating status or receiving input. These interfaces are not always traditional screens; they can be multisensory cues like indicator LEDs, tactile feedback, or audio alerts. **In effect, micro interactions are emerging at the part level, where**

each smart component provides a window into its ‘thinking’ or state.

Suppliers as strategic partners (Tier-0.5). As parts become smarter, we see OEMs rely on suppliers not just for hardware but for integrated solutions that include software, AI algorithms, and even user interface elements. This is driving a model where traditional Tier-1 suppliers evolve into ‘Tier-0.5’ partners—working long-term with OEMs on design and R&D, jointly developing subsystems that align with the OEM’s product vision. Such Tier-0.5 suppliers will increasingly deliver ready-made intelligent parts (or subsystems) that OEMs can plug-and-play into their products, dramatically cutting time-to-market. This partnership model, already seen in the automotive industry with digital cockpit platforms and ADAS systems, blurs the line between OEM and supplier in terms of innovation ownership.

From white-label products to white-label parts. A supplier’s part can come pre-packaged with intelligence and user experience, effectively serving as a mini-product that the end-user might never see, but feels through its functionality. A clear example of design thinking going into part development is AmpliSky—glass roofs for vehicles that switch from opaque or tinted to transparent in a very elegant manner, modulating light ambience and increasing comfort.

Growing electronics in every product

Today’s products—from toys to automobiles—contain far more electronics and software than ever before. The increasing percentage of electronics in product design is a fundamental driver of the shifts we are discussing.

Electronics are no longer add-ons; they define core product functionality in many cases.

Crucially, as more of the product value shifts to electronics and

embedded software, the expertise to create those elements often lies with specialised Tier-1 suppliers or design partners. An OEM of, say, home appliances may not be an expert in designing wireless modules or AI algorithms—they depend on an ecosystem of electronics providers. **The supplier’s role becomes larger and more collaborative,** since the innovation (and risk) tied up in that 35-50% electronic content must be managed in tandem with the OEM’s overall product development.

Modern vehicles are illustrative of this trend: packed with distributed electronics and software. In cars, what were once purely mechanical systems are now managed by dozens of electronic control units—a complexity driving new architectures to manage it all.

From central brains to edge intelligence

Hand in hand with the spread of electronics is a shift in where computing happens. We are moving away from a purely centralised model (one big brain in the device or the cloud) to a distributed, edge-centric model. In this model, each zone, module, or part of a system can have its own processing power and autonomy.

Automotive zonal architectures. A vivid illustration is the automotive industry’s transition to zonal E/E (electric-electronic) architectures. Traditionally, cars had dozens of independent ECUs, each dedicated to a single function (engine control, ABS, or infotainment). This was increasingly unwieldy as features grew.

The new approach is to place a small number of zone controllers across the vehicle’s physical regions (front, rear, and cabin), each handling multiple local functions. These zone controllers connect to a central high-performance computer for high-level decisions, while handling local sensor processing